Counter-controlled While Loop Handout

Problem Example 1 (Part 1)

Suppose each customer is allowed to purchase one item from a store. Write a program that prompts for and inputs the price of an item purchased by each of 5 customers and then calculates and displays the amount due by each customer after VAT is included.

```
Enter price for customer 1: 1200

Amount due by Customer 1 is R1 380,00

Enter price for customer 2: 950

Amount due by Customer 2 is R1 092,50

Enter price for customer 3: 500

Amount due by Customer 3 is R575,00

Enter price for customer 4: 2100

Amount due by Customer 4 is R2 415,00

Enter price for customer 5: 800

Amount due by Customer 5 is R920,00

Press any key to continue . . .
```

Problem Example 1 (Part 2)

Now revise your solution above and prompt for and input the number of customers. Your program must also calculate and display the total amount in rand value made by the store.

```
How many customers ? 3

Enter price for customer 1: 1500

Amount due by Customer 1 is R1 725,00

Enter price for customer 2: 2000

Amount due by Customer 2 is R2 300,00

Enter price for customer 3: 900

Amount due by Customer 3 is R1 035,00

Total made by the store: R5 060,00

Press any key to continue . . .
```

Problem Example 2

Accept as input the number of students in a class. Then prompt for and input the gender code of M or m for male or F or f for female, for each of the students. Note that any other code is deemed an error. Write a program that determines and displays the number of male students, the number of female students and the number of errors in the input.

```
How many students? 10

Enter gender code of M, m or F, f for all 10 students

Student 1: H
Student 2: f
Student 3: M
Student 4: F
Student 5: m
Student 6: v
Student 7: f
Student 8: m
Student 9: M
Student 10: M

Number of males          : 5
Number of females        : 3
Number of errors         : 2

Press any key to continue . . .
```